

QIAGEN

EZ2

Global Product Support EZ2 Installation Guideline IGL-5000-0003-001, QFS, QTS, EZ2



Installation Guideline

IGL-5000-0003-001, QFS, QTS, EZ2

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Contents

1	Introduction	5
1.1	General Information	5
1.2	EZ2 Installation Guideline Overview	5
1.3	Intended Use	5
1.3.1	Product Expected Use	5
1.3.2	Requirements for working with the product	6
2	Hazards and Precautions	7
2.1	Introduction	7
2.2	Safety	7
2.3	Proper Use	8
2.4	Electrical Safety	9
2.5	Biological Safety	11
2.6	Chemical Safety	12
2.7	Waste disposal	12
2.8	Mechanical Hazards	13
2.9	Heat Hazard	13
2.10	Radiation	14
2.11	Consumables	14
3	General Information	15
3.1	Instrument Names	15
3.2	Technology - Principle	15
3.3	Product principle	16
3.4	Instrument Overview	16
3.4.1	External features of the product	16
3.4.1.1	Touchscreen	17
3.4.1.2	Hood	17
3.4.1.3	Power Button	17
3.4.1.4	USB Ports	17
3.4.1.5	RJ-45 Ethernet port	17
3.4.1.6	Power Cord Socket	17
3.4.2	Internal features of the EZ2	17
3.4.2.1	Pipettor head	17
3.4.2.2	Worktable	18
3.4.2.3	Heating system	18
3.4.2.4	Tray	18
3.4.2.5	Magnet module	18
3.4.2.6	Camera	18
3.4.2.7	UV LED	19
3.4.2.8	Internal light	19
3.4.3	Product Specification	19
3.4.4	Consumables for Installation	19
4	Installation	20
4.1	System delivery and installation	20
4.1.1	EZ2 - Packing List	20
4.2	Installation of the EZ2	22
4.2.1	Requirements	22
4.2.1.1	Site	22

4.2.1.2	Environment	22
4.2.1.3	Power	22
4.2.1.4	Grounding	22
4.2.2	Unpacking The EZ2 Connect	23
4.2.2.1	Power on the Instrument	27
4.2.2.2	Software Configuration	27
4.2.2.3	Network Connections	30
4.2.3	Camera Exposure Calibration	31
4.2.4	Workflow Installation Test	32
4.2.5	After Run Maintenance	38
4.2.6	Daily Maintenance	39
4.2.7	UV Test	40
4.2.8	Installation Completion	42

1 Introduction

1.1 General Information

QIAGEN Global Product support (GPS) is responsible for the creation and maintenance of Instrument Service documentation. It is the policy of QIAGEN to improve products as new techniques and components become available. QIAGEN reserves the right to change specifications at any time. In an effort to produce useful and appropriate documentation, we welcome your comments and suggestions on this publication.

QIAGEN field specialists can contact GPS via e-mail (service@qiagen.com) or via phone.

Specialists outside QIAGEN can contact the appropriate Technical Services department.

INFORMATION 	If an error is found in this installation guideline please contact Global Product Support immediately.
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1.2 EZ2 Installation Guideline Overview

This Installation Guideline represents an integral part of the QIAGEN EZ2 platform.

This guideline is not intended for the service and repair. This installation guideline provides detailed information regarding the installation, operation, and testing of the EZ2.

The instructions contained in the installation guideline are to be strictly observed. Pay particular attention to any safety information. The information is presented in the following chapters:

1. Introduction (this chapter)
2. Hazards and Precautions
3. General Information
4. Installation

1.3 Intended Use

1.3.1 Product Expected Use

The EZ2 Connect and EZ2 Connect Fx systems are designed to perform automated isolation and purification of nucleic acids. The EZ2 is intended to be used only in combination with QIAGEN kits indicated for use with the EZ2 instrument for the applications described in the kit handbooks. The EZ2 system is intended for use by professional operators, such as technicians and physicians trained in molecular biology techniques and the operation of the EZ2 system.

The EZ2 Connect MDx system is designed to perform automated isolation and purification of nucleic acids in molecular diagnostic and/or molecular biology applications. The EZ2 Connect MDx is intended to be used only in combination with QIAGEN kits indicated for use with the EZ2 Connect MDx instrument for the applications described in the kit handbooks. The EZ2 Connect MDx system is intended for use by

professional operators, such as technicians and physicians trained in molecular biological techniques and the operation of the EZ2 Connect MDx system.

INFORMATION 	Refer to the relevant User Manual for the EZ2 system for the appropriate intended use statements.
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1.3.2 Requirements for working with the product

This section covers the general level of competence and training necessary for installation, use, daily maintenance, cleaning, preventive maintenance and servicing of the product. Appropriate training for all levels can be arranged with QIAGEN.

Task	Personnel	Knowledge
Delivery	No special requirements	No special requirements
Installation	Laboratory Technicians or equivalent	Appropriately trained and experienced personnel, familiar with use of computers and automation in general.
Routine Use (running protocols)	Laboratory technicians or equivalent	Professional users, such as technicians or physicians, trained in molecular biology techniques.
Routine Maintenance	Laboratory technicians or equivalent	Professional users, such as technicians or physicians, trained in molecular biology techniques.
Servicing and Annual Preventative Maintenance	QIAGEN authorized specialists	Trained, certified and authorized by QIAGEN.

Table 1.3-1 Requirements for EZ2 users

2 Hazards and Precautions

2.1 Introduction

Before using or servicing the product, it is essential that you read this Installation Guideline as well as the current and appropriate User Manual carefully and pay particular attention to the safety information. The handbooks contain information and warnings that must be followed by the user and field service personnel to ensure safe operation of the instrument and to maintain the instrument in a safe condition.

Possible hazards that could harm the user or result in damage to the instrument are clearly stated at the appropriate places throughout this manual.

The following safety conventions will be used throughout this manual.

WARNING 	The term WARNING is used to inform you about situation that could result in personal injury to you or other persons. Details about these circumstances are given in a box like this one.
CAUTION 	The term CAUTION is used to inform you about situations that could result in damage to the instrument or other equipment. Details about these circumstances are given in a box like this one.
INFORMATION 	The term INFORMATION is used to inform you about hints and tricks . Details about these circumstances are given in a box like this one.

The advice given in this guideline is intended to supplement, not supersede, the normal safety requirements prevailing in the user's country.

2.2 Safety

WARNING 	You must observe all safety requirements when installing and operating the product. Failure to install or use the system in a manner specified by QIAGEN may result in personal injury and / or equipment damage .
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The product shall be installed in accordance with the Installation in Chapter 4 of this service manual.

The mains plugs shall be easily accessible in case of the need of quickly disconnecting the equipment from the mains power.

The product shall be used in accordance with the instructions provided through dialogs appearing on the touchscreen, the manuals, instructions for use, and by QIAGEN Technical Service, and within limits set by the technical specifications.

Follow all generally accepted laboratory safety procedures and applicable laws and regulations. Use appropriate safety precautions when handling biological material.

2.3 Proper Use

WARNING 	RISK OF PERSONAL INJURY AND MATERIAL DAMAGE Improper use of the EZ2 may cause personal injuries or damage to the system. The EZ2 must only be operated by qualified personnel who have been appropriately trained. Servicing of the EZ2 must only be performed by QIAGEN authorized Field Service Specialists.
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- Use only QIAGEN components, otherwise any right to make a claim under the guarantee may be invalidated.
- Perform the maintenance as described in Maintenance Chapter of the relevant User Manual.
- QIAGEN will charge for repairs that prove to be required due to incorrect maintenance.

WARNING 	RISK OF PERSONAL INJURY AND MATERIAL DAMAGE The EZ2 is too heavy to be lifted by one person. To avoid personal injury or damage to the instrument, do not lift the instrument alone. Use the handle attached to the box to lift the EZ2. After the EZ2 has been unpacked, two people must lift the instrument. Lift the instrument by placing your hands underneath the bottom of the instrument.
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WARNING 	RISK OF PERSONAL INJURY AND MATERIAL DAMAGE Do not attempt to move the EZ2 during operation.
--	---

In case of emergency, power OFF the EZ2 at the power switch located in the front of the instrument and unplug the power cord from the power outlet.

CAUTION 	Damage to the instrument Avoid spilling water or chemicals onto the EZ2.
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CAUTION 	Damage to the instrument Do not use spray bottles containing alcohol or disinfectant to clean surfaces of the EZ2 instrument. Spray bottles should be used only to clean items that have been removed from the worktable and if permitted by local laboratory operating practices.
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WARNING 	Risk of fire or explosion When using ethanol or alcohol-based liquids on the EZ2, handle such liquids carefully and in accordance with the required safety regulations. If liquid has been spilled, wipe it off and leave the EZ2 hood open to allow flammable vapors to disperse.
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CAUTION 	Damage to the instrument After wiping the worktable with paper towels, make sure that no residual pieces of paper towel remain. Pieces of paper towel remaining on the worktable could lead to a worktable collision.
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WARNING 	Risk of Explosion The EZ2 is intended for use with reagents and substances supplied with QIAGEN kits as outlined in respective Information for use. Use of other reagents and substances may lead to fire or explosion.
---	--

If hazardous material is spilled on or inside the EZ2, the user is responsible for carrying out appropriate decontamination.

Note: Do not place items on top of the EZ2.

CAUTION 	Damage to the instrument Ensure that the EZ2 is switched off before you manually move the mechanical components of the instrument.
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CAUTION 	Damage to the instrument Do not lean against the instrument or touchscreen.
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2.4 Electrical Safety

Note: If operation of the instrument is interrupted in any way (for example, due to interruption of the power supply or a mechanical error), first switch off the EZ2 instrument, then disconnect the electrical cord from the power supply before attempting troubleshooting or service activity.

Disconnect the line power cords from the power outlet before servicing.

WARNING 	ELECTRICAL HAZARD Disconnect the line power cords from the power outlet before servicing. Lethal voltages inside the instrument When the instrument is connected to line power, terminals may be live, and opening covers or removing parts is likely to expose live parts.
WARNING 	ELECTRICAL HAZARD Any interruption of the protective conductor (earth/ground lead) inside or outside the instrument or disconnection of the protective conductor terminal is likely to make the instrument dangerous. Intentional interruption is prohibited.

WARNING 	Damage to electronics Before powering ON the instrument, make sure that the correct supply voltage is used. Use of incorrect supply voltage may damage the electronics. To check the recommended supply voltage, refer to the specifications indicated in the type plate of the instrument.
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WARNING 	Risk of electric shock Do not open any panels on the EZ2. Risk of personal injury and material damage Only perform maintenance that is specifically described in this user manual, or service manual.
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To ensure satisfactory and safe operation of the EZ2, follow the advice below:

- Make sure the line power cord is connected to a line power outlet that has a protective conductor (earth/ground).

INFORMATION 	The ground prong on the mains plugs must not be removed and the plugs should only be connected to grounded outlets as per local and national regulations. Use of power supplies that disable the grounding connections to the instrument is dangerous and prohibited.
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- Keep mains plugs easily accessible in case the equipment needs to be disconnected quickly from the mains power.
- Use only power supplies and cords supplied with the system.
- Power supplies and cords should be inspected periodically and replaced if damaged or altered.
- Do not attempt to make any internal adjustments or replacements.
- Do not operate the instrument with any covers or parts removed.
- If water, liquids or reagents have spilled inside the instrument, switch off the instrument and disconnect it from the line power supply. Contact Global Product Support.

INFORMATION 	Note that the power switch does not disconnect the equipment completely from mains power.
---	--

- Servicing must be performed by trained QIAGEN service engineers.
- If the instrument becomes electrically unsafe for use, make the instrument inoperative and secure it against unauthorized or unintentional operation. Contact Global Product Support.

The instrument is likely to be electrically unsafe when it:

- The EZ2 or the line power cord appear to be damaged.
- The EZ2 has been stored in unfavorable conditions for a prolonged period.
- The EZ2 has been subjected to severe transport stresses.
- Liquids have come into direct contact with electrical components of the EZ2.
- The power cord has been exchanged with a non-official power cord.
- Shows visible damage.
- The EZ2 has been stored in unfavorable conditions for a prolonged period.

WARNING 	Electric hazard Do not touch the EZ2 with wet hands.
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WARNING 	Electric hazard Never install a fuse different from that specified in the service manual.
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2.5 Biological Safety

The EZ2 instruments are used in combination with Biological material.

WARNING 	Always clean and decontaminate the instrument prior to service or use.
WARNING 	Samples containing infectious agents Some samples used with the EZ2 instrument may contain infectious agents. Handle such samples with the greatest of care and in accordance with the required safety regulations. Always wear safety glasses, gloves, and a lab coat. The responsible body (for example, a laboratory manager) must take the necessary precautions to ensure that the surrounding workplace is safe, and that the instrument operators are suitably trained and not exposed to hazardous levels of infectious agents as defined in the applicable Material Safety Data Sheets (MSDSs) or the OSHA^{1,*} ACGIH,[†] or COSHH[‡] documents. Venting for fumes and disposal of wastes must be in accordance with all national, state, and local health and safety regulations and laws.

When handling biological material, use safe laboratory procedures as outlined in publications such as Biosafety in Microbiological and Biomedical Laboratories, HHS (HHS Publication Number) (<https://www.cdc.gov/labs/pdf/CDC-BiosafetymicrobiologicalBiomedicalLaboratories-2009-P.pdf>).

* OSHA: Occupational Safety and Health Administration (United States of America).

† ACGIH: American Conference of Government Industrial Hygienists (United States of America).

‡ COSHH: Control of Substances Hazardous to Health (United Kingdom).

2.6 Chemical Safety

WARNING 	Hazardous chemicals Some chemicals used with the EZ2 instrument may be hazardous or may become hazardous after completion of a purification. Always wear safety glasses, gloves, and a lab coat. The responsible body (e.g., laboratory manager) must take the necessary precautions to ensure that the surrounding workplace is safe, and that the operators are suitably trained and not exposed to hazardous levels of infectious agents as defined in the applicable Material Safety Data Sheets (MSDSs) or OSHA*, ACGIH†, or COSHH‡ documents. For more information, visit: www.qiagen.com/support/msds.aspx. Venting for fumes and disposal of wastes must be in accordance with all national, state, and local health and safety regulations and laws.
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* OSHA: Occupational Safety and Health Administration (United States of America).

† ACGIH: American Conference of Government Industrial Hygienists (United States of America).

‡ COSHH: Control of Substances Hazardous to Health (United Kingdom).

WARNING 	Toxic Fumes Do not use bleach to clean or disinfect the EZ2 instrument. Bleach in contact with salts from the buffers can produce toxic fumes.
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WARNING 	Toxic Fumes Do not use bleach to disinfect used labware. Bleach in contact with salts from the buffers can produce toxic fumes.
---	--

Note: If you work with volatile solvents, toxic substances, etc., you must provide an efficient laboratory ventilation system to remove vapors that may be produced.

2.7 Waste disposal

Used consumables, such as reagent cartridges and disposable filter-tips, may contain hazardous chemicals or infectious agents from the purification process. Such waste must be collected and disposed of properly, according to local safety regulations.

For more information about how to dispose of the EZ2, see the User Manual Appendix A: Waste Electrical and Electronic Equipment (WEEE).

CAUTION 	Hazardous chemicals and infectious agents Waste may contain toxic or infectious material and must be disposed of properly. Refer to your local safety regulations for proper disposal procedures.
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2.8 Mechanical Hazards

The hood of the EZ2 Instrument must remain closed during user operation of the instrument. The hood has a lock to prevent opening during runs.

CAUTION 	Moving parts A service switch allows the instrument to be operated with the hood open when the rear cover is removed. Use caution when working with the instrument and the covers are removed.
---	---

The worktable of the EZ2 instrument moves during operation of the instrument. While loading the worktable, always stand clear of the instrument. Do not lean on the worktable when the robotic arm of the instrument is moving to reach loading position with its lid open. Wait until the robotic arm completes its movements before you start to load or unload.

WARNING 	Moving parts To avoid contact with moving parts during operation of the EZ2, the instrument must be operated with the lid closed. If there is a problem with the EZ2, great care should be taken if the instrument is operated with covers off during service.
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WARNING 	Be careful when cleaning the piercing unit - the spikes are very sharp and can be contaminated .
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WARNING 	Moving parts Avoid contact with moving parts during operation of the EZ2. Under no circumstances should hands be placed under the pipetting arm during movement. Do not attempt to remove any plasticware from the workdeck whilst the instrument is operating.
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2.9 Heat Hazard

The EZ2 worktable contains a heating system.

WARNING 	Hot surface The heating system can reach temperatures of up to 95°C (203°F). Avoid touching the heating system when it is hot.
CAUTION 	Risk of overheating To ensure proper ventilation, maintain a minimum clearance of 15 cm (5.9 in.) at the sides and rear of the EZ2. Slits and openings that ensure the ventilation of the EZ2 must not be covered.

2.10 Radiation

The EZ2 instrument has a UV LED. The wavelength of the UV light produced by the UV LED is 270 to 285 nm. This wavelength corresponds to ultraviolet light type C, which can be used for decontamination procedures. A mechanical lock ensures that the hood must be closed for operation of the UV LED.

WARNING 	UV radiation Avoid looking directly into UV light. Do not expose your skin to UV light. Great care should be taken if the instrument is operated with covers off during service.
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The EZ2 instrument has a 2D handheld barcode scanner to allow kit bar code and sample barcode scanning.

WARNING 	Risk of personal injury Hazard level 2 laser light: Do not stare into the light beam when using the handheld barcode scanner.
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2.11 Consumables

CAUTION 	Unsupported Consumables Do not connect or use any consumable, accessories, or external equipment other than those specified.
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3 General Information

3.1 Instrument Names

The EZ2 system is available in three different variants: EZ2 Connect, EZ2 Connect Fx and EZ2 Connect MDx. The EZ2 Connect is intended for general life sciences applications. The EZ2 Connect Fx is intended for forensic, human identification applications. The EZ2 Connect MDx includes the regulated protocols for diagnostic applications. Use of the term EZ2 refers to all three variants, unless otherwise specified.

3.2 Technology - Principle

The EZ2 Connect performs fully automated nucleic acid purification from up to 24 samples in molecular diagnostic and molecular biology applications using magnetic particle technology.

The EZ2 Connect has preinstalled protocols that are used with QIAGEN kits to purify nucleic acids. The instrument's touchscreen display allows the user to easily select protocols. The intuitive software/user interface guides the user through the run setup process, which includes selecting variable parameters. Afterward, the operator loads labware and samples onto the EZ2 Connect worktable, while following the instructions shown on the display. The order and contents of the labware are determined by each protocol. When the protocol run starts, the isolation of nucleic acids is carried out automatically in a modular workflow. The workflow includes sample lysis, nucleic acid binding to magnetic particles, contaminant removal by washing, and, finally, elution from magnetic beads. The execution of individual steps varies and is optimized for each protocol.

The aspiration and dispensation of samples and reagents and the separation of magnetic particles are performed by the 24-channel pipettor head. If required by the protocol, the temperature of the liquids is controlled by the heating system.

The EZ2 Connect Fx incorporates all the characteristics of the EZ2 Connect, as well as some additional features, which are particularly relevant for forensic workflows. These features include:

- An internal camera, which is used for load checks and bar code reading
- An external bar code reader, which is used for reading sample IDs and kit bar codes
- A sample recovery function for forensic protocols
- Extended user management
- Extended User Interface
- Additional reporting functions
- UV schedule function

The EZ2 Connect MDx performs fully automated nucleic acid purification from up to 24 samples per single run in molecular diagnostic and molecular biology applications using magnetic particle technology. The EZ2 Connect MDx is designed to automate EZ1 DSP Kits available from QIAGEN, processing of non-DSP kits is also supported.

The EZ2 Connect MDx gives the option to start a protocol either in the IVD mode of the software (for validated IVD applications only) or in the Research mode of the software (for Molecular Biology Applications (MBA) only). The usage of IVD protocols is only possible and strictly restricted to the IVD mode of the software. This User Manual focuses on operating the EZ2 Connect MDx in the IVD-software mode. For detailed instructions on how to operate the EZ2 Connect MDx using the Research mode of the software (with MBA-protocols or any customized protocols), please refer to the User Manual of the EZ2 Connect MDx.

EZ2 Connect MDx is preinstalled with various protocols for processing QIAGEN EZ1 kits for purification of RNA, genomic DNA, and viral and bacterial nucleic acids, and RNA. The user must first log in to the software in IVD mode using the touchscreen, then scans a kit barcode and loads labware, samples, and reagents onto the EZ2 Connect MDx worktable. The user then closes the instrument hood and starts the protocol, which provides all necessary commands for sample lysis and purification. A fully automated load check helps to ensure correct loading of the worktable.

Users can stay connected to their instrument through the built-in screen and also remotely with a computer or a mobile device (e.g., a tablet) and the QIA sphere App using the associated QIA sphere Connectivity Package.

The aspiration and dispensation of samples and reagents and the separation of magnetic particles are performed by the 24-channel pipettor head. If required by the protocol, the temperature of the liquids is controlled by the heating system.

The EZ2 Connect MDx incorporates the following features:

- An internal camera, which is used for load checks and bar code reading
- An external bar code reader, which is used for reading sample IDs and kit bar codes
- Extended user management
- Extended User Interface
- Additional reporting functions
- UV schedule function

3.3 Product principle

The EZ2 Connect MDx system is designed to perform automated isolation and purification of nucleic acids in molecular diagnostic and/or molecular biology applications.

One of the following QIAGEN kits is required to perform automated isolation and purification of nucleic acids on the EZ2 Connect MDx in the IVD mode of the software:

- EZ1&2 DSP DNA Blood Kit (Part Number 62124)
- EZ1&2 DSP Virus Kit (Part Number 62724)

The research mode of the software supports the usage of other EZ1&2 or EZ2 kits for life science applications.

3.4 Instrument Overview

An introduction to the features of the instrument is provided here. For further information, refer to the relevant User Manual.

3.4.1 External features of the product



Picture 3.4-1 The EZ2 Connect instrument

3.4.1.1 Touchscreen

The EZ2 has a 10.1-inch color touchscreen display with a resolution of 1280 x 800 pixels. The Graphical User Interface (GUI) is displayed on the touchscreen, allowing the user to completely operate the instrument.

3.4.1.2 Hood

The EZ2 hood protects the interior of the instrument from external contamination during protocol runs. Additionally, the hood protects operators from UV radiation during decontamination procedures.

The hood must be closed to enable a protocol run to start. The hood is locked at the beginning of a run, and remains locked throughout the duration of the run. This protects users from moving parts on the worktable. The hood can be manually opened to gain access to the worktable when no protocol is running.

3.4.1.3 Power Button

The power button is located on the front bottom right corner of the EZ2. Press the button to turn on the EZ2. After the button is pressed, the power button lights up, the startup screen appears on the touchscreen, and the instrument initializes.

To conserve energy, the EZ2 can be turned off when not in use. To turn off the EZ2, press the power button.

3.4.1.4 USB Ports

The EZ2 has 3 USB ports. One is located on the front of the instrument next to the power button. Two are located on the upper back of the touchscreen display.

The USB ports allow you to connect a USB drive to the EZ2, in order to, for example, transfer report files.

In addition, the USB drive can be used to upload protocols, or update the software.

The bar code scanner is connected to the EZ2 using any one of the 3 available USB ports.

Important: Use only the USB flash drive provided by QIAGEN. Do not connect other USB flash drive devices to USB ports

Important: Do not remove the USB drive while downloading or transferring data or software to or from the instrument.

Important: Always turn off the EZ2 in order to plug or unplug the Wi-Fi USB device. Plug-and-play of the Wi-Fi USB device while the instrument is turned on is not supported.

3.4.1.5 RJ-45 Ethernet port

The RJ-45 Ethernet port is located on the back of the instrument near the power connection. The Ethernet connection is 10/100 BASE T.

3.4.1.6 Power Cord Socket

The power cord socket is located at the back of the EZ2.

3.4.2 Internal features of the EZ2

3.4.2.1 Pipettor head

The pipettor head is mounted above the worktable and moves in the Z-direction (meaning up and down) to reach the sample and reagent tubes on the worktable. The worktable itself moves in the Y-direction (meaning front to back) so that the pipettor head is above the appropriate position in the cartridge or tip rack during each action that is performed by the instrument.

The pipettor head contains 24 high-precision syringe pumps that are connected to tip adapters which can be attached to filter-tips. The syringe pumps operate simultaneously and can aspirate or dispense small volumes of liquid (20–1000 µl) through the attached filter-tips.

Another component of the pipettor head is the piercing unit that is located behind the tip adapters. The piercing unit is a row of 24 metal spikes that puncture the foil sealing the reagent cartridges. During operation, the piercing unit, controlled by the EZ2, opens all wells of the reagent cartridges in a dedicated order. The pipettor head then automatically picks up filter-tips from the tip rack and performs aspiration and dispensation operations at different locations on the worktable before ejecting the tips back into the tip rack at the end of the run.

3.4.2.2 Worktable

The EZ2 worktable contains two movable racks that hold all the labware required for a protocol run, and the heating system, which controls the temperature of samples during a run.

The cartridge rack consists of two separate parts. The left cartridge rack is used for cartridges in positions from 1 to 12. The right cartridge rack is used for cartridges in positions 13 to 24. On the worktable, the cartridge rack is located behind the tip rack. It holds up to 24 reagent cartridges.

Sealed reagent cartridges are prefilled and contain the reagents needed for a protocol run. Each cartridge consists of 10 sealed reagent wells and 2 empty heating positions. One heating position is a well and the other is a slot that can hold a tube.

The EZ2 tip rack consists of two separate parts. The left part of the tip rack is used for labware in positions from 1 to 12. The right part of the tip rack is used for labware in positions 13 to 24.

The tip racks are located at the front of the worktable. Each consists of four rows and 12 positions:

- Row A — which is the row that is closest to the cartridge rack, holds up to 24 sample tubes.
- Important: Only use QIAGEN recommended tubes (see respective EZ1&2 and EZ2 kit handbooks for more information).
- Row B — usually holds up to 24 tip holders with filter tips, which are provided with the EZ1/EZ2 kits (see respective EZ1&2 kit handbook for more information).
- Row C — may contain a tube with manually filled components, such as carrier RNA or Ethanol (see respective EZ1&2 and EZ2 kit handbooks for more information).
- Row D -- usually contains the elution tube.

3.4.2.3 Heating system

The heating system is located under the rear part of cartridge rack. If needed, it heats up the cartridge rack wells in positions 11 and 12.

3.4.2.4 Tray

The tray is located underneath the tip and cartridge racks. Its role is to prevent contamination of the EZ2 that could be caused by liquids that are unintentionally spilled.

3.4.2.5 Magnet module

The EZ2 magnet module consists of magnets that are used to capture magnetic particles that are present in the liquid aspirated into the filter-tips.

3.4.2.6 Camera

The EZ2 Connect Fx and EZ2 Connect MDx have built-in cameras that are used for load checks and reading bar codes from cartridges.

Load checks are done before a protocol run starts. The camera checks if all the labware has been loaded in the correct positions. The results of the load check are shown on the screen. For more information on the load check.

The camera also reads the cartridge 2D bar codes. Information that is collected from the bar codes is included in run reports.

There is no internal camera on the EZ2 Connect.

3.4.2.7 UV LED

The EZ2 is equipped with a UV LED for decontamination. During the maintenance decontamination procedure, the UV LED moves over the worktable.

Note: The hood must be closed before starting a maintenance procedure and is automatically locked during the procedure.

3.4.2.8 Internal light

The EZ2 has a built-in LED light strip. The internal light strip illuminates the worktable and informs the user about the current status of the run. There are two modes of the LED light:

Blinking light – indicates that operator activity is required (for example, when an error occurs).

Constant light – the default setting, used in all other situations.

3.4.3 Product Specification

Refer to the current relevant User Manual for the EZ2 system for the Product Specifications and Technical Details.

3.4.4 Consumables for Installation

SAP #	Description	Comment
n/a	EZ2 Kit	Any kit that can be run on the appropriate EZ2 instrument to be installed. This could be the kit the client will be using, an example Kit, or a Service Kit (if available).
9017734	Plasticware kit	Optional. Plasticware kit containing empty reagent cartridges, tips and tip holders, and the 1.5 and 2mL tubes.

Table 3.4-1 Consumables for installation

4 Installation

4.1 System delivery and installation

The unpacking and installation of the product can be completed by the client. An installation service can be purchased where a QIAGEN authorized installer/specialist performs the installation of the EZ2 system.

4.1.1 EZ2 - Packing List

Parts			Quantity
Part #	Part	Picture	
9003210	EZ2 Connect		1
9003220	EZ2 Connect Fx		
9003230	EZ2 Connect MDx		
N/A	Quick Start Guide(s) for appropriate instrument: • HB-2924 MDx QSG EZ2 Connect • HB-2925 QSG EZ2 Connect		1
9027012	Cartridge Rack (Left)		1
	Cartridge Rack (Right)		1
9027009	Tip Rack (Left)		1
	Tip Rack (Right)		1

Table 4.1-1 EZ2 Packing List

Parts			Quantity
Part #	Part	Picture	
9027010	Flip Cap Rack (Left) <i>Only included with the Fx Instruments</i>		1 <i>Only included with the Fx</i>
	Flip Cap Rack (Right) <i>Only included with the Fx Instruments</i>		1 <i>Only included with the Fx</i>
9027102	Silicone grease		1
	Certificate of Manufacture EZ2 Connect		1
9026073	Handheld Barcode Reader <i>Only included with the MDx and FX Instruments</i>		1
9017850	USB Stick		1
9015126	Power Cord Set, International		1

Table 4.1-1 EZ2 Packing List

4.2 Installation of the EZ2

WARNING 	Risk of system damage If the EZ2 is transported in temperatures below 0° C (32° F), it is recommended to wait 24 hours before switching on the instrument.
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4.2.1 Requirements

4.2.1.1 Site

The EZ2 is recommended be installed in a normal laboratory environment (18°C to 30°C and 10 - 75% relative humidity) for optimal performance. Refer to [“Product Specification” on page 19](#) for more instrument specific information.

The EZ2 must be located out of direct sunlight, away from heat sources, and away from sources of vibration and electrical interference. The site of installation should also be level, dry, clean and free of excessive drafts, excessive moisture, excessive dust, and not subject to large temperature fluctuations.

Allow at least 10 cm (4.9 in.) of free space behind and to the sides of the instrument for cooling and cabling.

The EZ2 must be placed within approximately 1.5 m (59 in.) of a properly grounded (earthed) AC power outlet. The power line to the product should be voltage regulated and surge protected.

INFORMATION 	It is recommended to plug the instrument directly into its own power outlet and not to share the power outlets with other lab equipment.
CAUTION 	Risk of overheating To ensure proper ventilation, maintain a minimum clearance of 10 cm (4.9 in.) at the sides and rear of the product. Slits and openings that ensure the ventilation of the EZ2 must not be covered.

4.2.1.2 Environment

- Temperature: 18°C to 30°C
- Humidity: 10% to 75% relative humidity
- Draft-free location not close to a window
- Keep the instrument out of direct sunlight
- Altitude: Up to 2000 m (6500 ft.)

4.2.1.3 Power

The product operates at:

- 100–240 V AC, 50–60 Hz, 1000VA

Ensure that the voltage rating of the product is compatible with the AC voltage available at the installation site. Mains supply voltage fluctuations are not to exceed 10% of nominal supply voltages.

4.2.1.4 Grounding

To protect operating personnel, the product must be correctly grounded (earthed). The product is equipped with a 3-conductor AC power cord. To preserve this protection feature, do not operate the product from AC power outlet that has no ground (earth) connection.

4.2.2 Unpacking The EZ2 Connect

The instrument is delivered with all the necessary components for setting up and running the instrument. The package also contains a packing list and quick start guide. It is recommended to check the packing list to ensure all items are present and undamaged.



Picture 4.2-1 EZ2 Shipping Crate

1. Before unpacking the EZ2 move the package to the site of installation and check that the package does not have damage. If damage is noted, notify the local service manager and confirm installation before continuing.
2. Remove the straps securing the crate to the shipping pallet.
3. Cut the tape on the top of the box and open the flaps and reveal the accessories box.



Picture 4.2-2 Opened EZ2 shipping box and accessories box (external box already removed)

4. Remove the accessories box and set aside. Check that all items were delivered and are not damaged.



Picture 4.2-3 EZ2 unpacking accessories

5. To remove the exterior cardboard box, use the handles on the sides of the box to lift the box straight up and over the contents to remove the exterior cardboard box.



Picture 4.2-4 Use handles to lift the box

6. Remove the two pieces of foam from atop the instrument.



Picture 4.2-5 Remove foam from atop the instrument

7. Remove the plastic covering from the instrument.



Picture 4.2-6 Plastic covering removal

WARNING 	Risk of Injury Utilize at least two individuals when lifting the EZ2 instrument. When lifting, bend at the knees and keep your back straight.	
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8. Remove the clear plastic covering from the instrument.
9. Check the EZ2 system for damage or defects. If any damage or defects are noted, contact the local service manager to confirm continued installation.
10. Lift under both the left and right sides of the instrument and place the instrument into position on the bench.

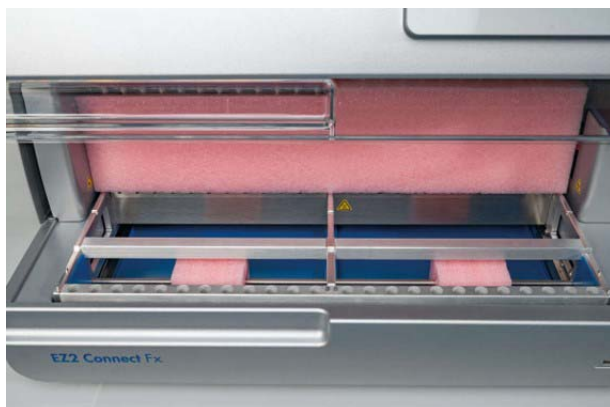
WARNING 	Risk of material damage Do not lift the EZ2 by holding the touchscreen or hood, as this may damage the instrument.	
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11. Lift open the clear cover hood to access the inside of the EZ2 instrument.



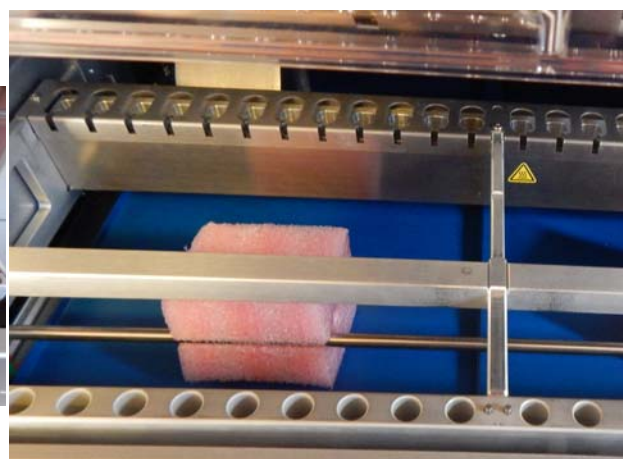
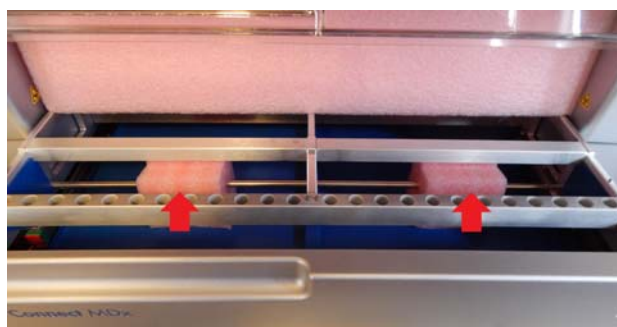
Picture 4.2-7 EZ2 with hood closed

12. Remove the z- and p-axis transport lock foam by pulling the bottom of the large piece of foam toward the front of the instrument.



Picture 4.2-8 Foam removal Z-P axis

13. Remove the small Y-foam blocks from the worktable by rotating until the split in the foam is toward the front and then pushing the bottom of the transport lock toward the rear direction. Remove the transport lock from the back.



Picture 4.2-9 Y Axis foam removal

14. Remove the silica pack from the unit.
15. Carefully remove the protective film from the touchscreen.
16. Insert the USB Wi-Fi adapter (if available) into the USB port above and behind the touchscreen.



Picture 4.2-10 Touchscreen USB ports, as viewed from the back of the instrument

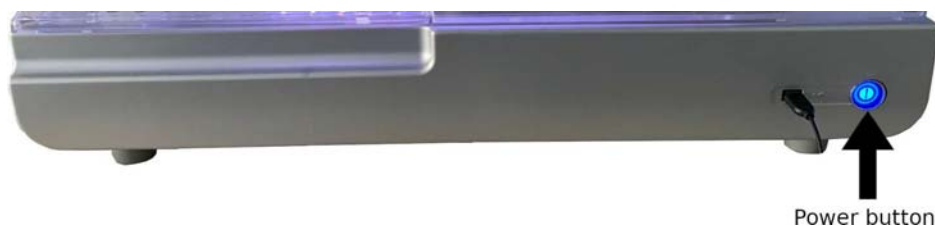
17. Connect the handheld barcode reader, if applicable, to an available USB port.



Picture 4.2-11 Available USB Ports

4.2.2.1 Power on the Instrument

1. Connect the appropriate power cord to the power inlet on the instrument and connect the power cord to the wall outlet.
2. Ensure the hood is closed.
3. Ensure that the EZ2 has equilibrated to ambient temperature before operating it.
4. Before powering on the instrument, the Barcode Reader (if applicable) and WiFi adapter (if applicable) must be installed. Confirm these are installed, if applicable.
5. Press the power button on the front of the instrument to power on the instrument.



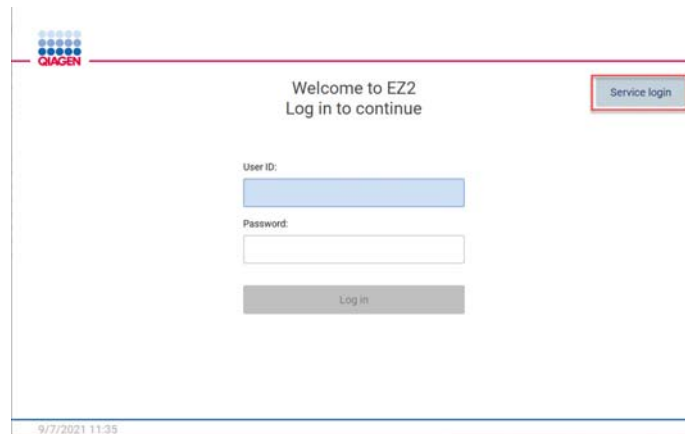
Picture 4.2-1 Location of the power button

6. Observe the instrument during the boot process: the power button will light up, the startup screen will appear on the touchscreen, and the instrument will initialize.
7. The EZ2 must initialize and must display the log on screen successfully.

4.2.2.2 Software Configuration

The EZ2 is operated with the touchscreen, which requires a log in. During installation, all actions should be possible using the user log ins. However, in order to access the service functions, the use of a service key and unique password is required.

1. Log into the EZ2 GUI.
 - 1.1. To log in as a service user:
 - Insert the USB drive, with the service file in the root of the USB drive, into an available EZ2 USB port.
 - Press the **Service login** button in the upper right of the screen.



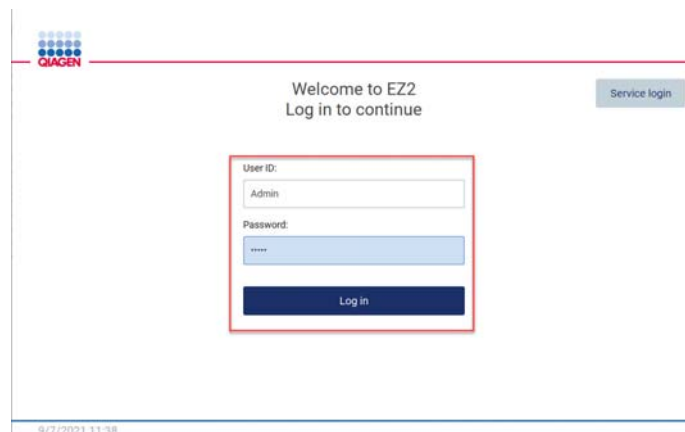
Picture 4.2-1 EZ2 Service Log In Screen

- Enter your specific EZ2 service password in the password prompt window.
- After entering the password, press **Accept**. If the password is accepted, the software will unlock.



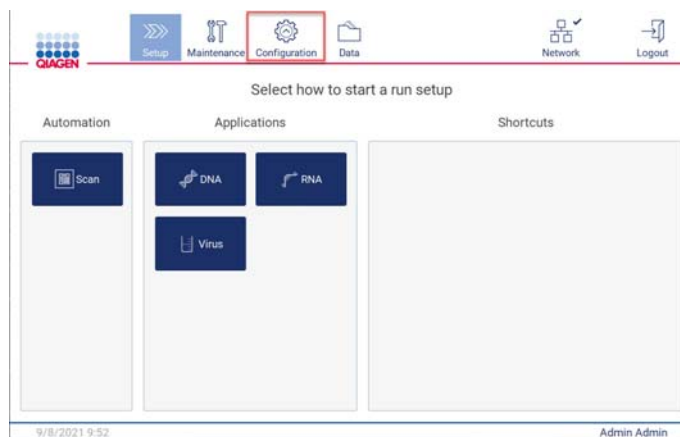
Picture 4.2-2 Enter the service password

- Read the UV warning and respond appropriately.
- 1.2. For log in as a user, the default log in information is:
- User ID: Admin
 - Password: Admin
 - Press **Log in** after entering the appropriate information.



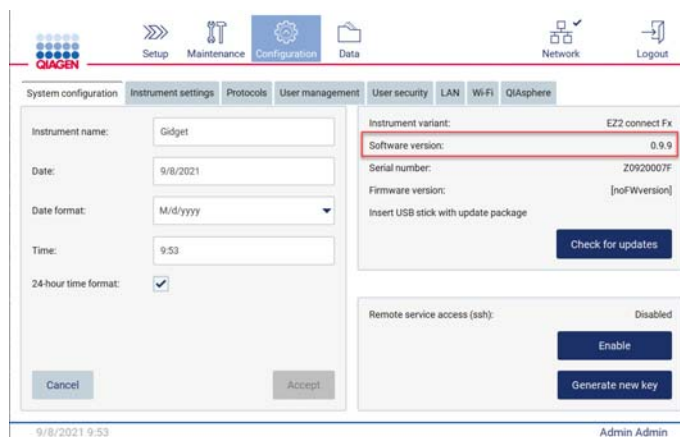
Picture 4.2-3 EZ2 Log in default user information

- Press the **Configuration** environment icon.



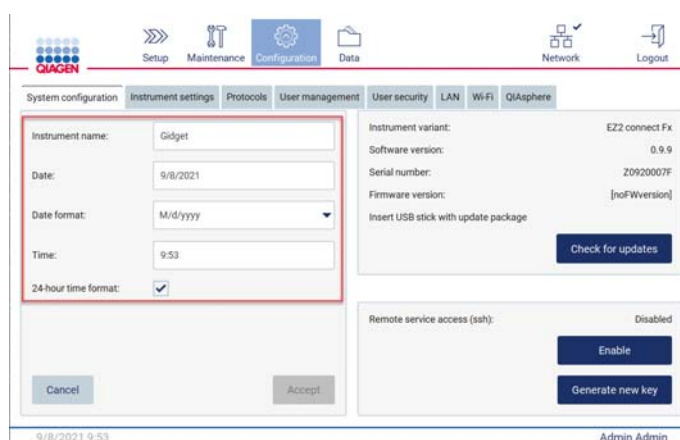
Picture 4.2-4 Configuration Environment Icon

- Check the latest version of the User Software is installed. Update the software if not on the latest software. If the software can not be updated to the latest version, document the reason in the Service Report.



Picture 4.2-5 Software Version

- Set the system date, date format, and time by touching the areas on the screen and adjusting the settings to reflect the current date, time and preferred date format and hour format.



Picture 4.2-6 Update instrument configuration

- If necessary, change the device name.
- After changing the settings, click **Accept** to save the settings.

4.2.2.3 Network Connections

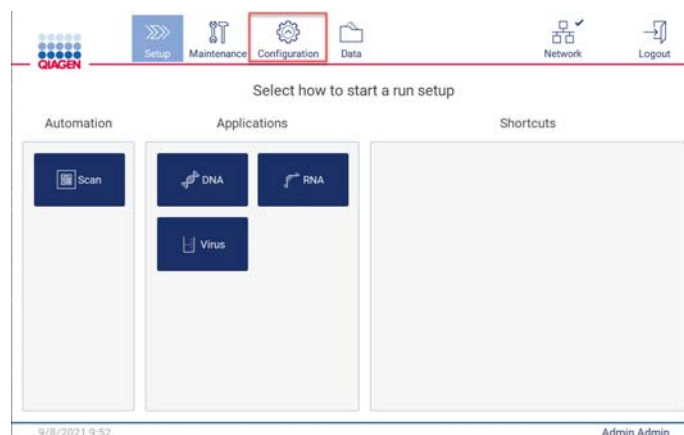
The EZ2 Connect series of instruments will have the option to be connected to a network for file transfer and instrument status monitoring via QIASphere. This network connection can be made either with the Wireless Network (Wi-Fi) or with a wired network. This is an optional feature, so the following is only if the instrument will be connected to the network.

1. Connect the Ethernet cable to the designated network outlet in the lab, if applicable, or confirm the Wi-Fi adapter is inserted into one of the touchscreen USB ports.

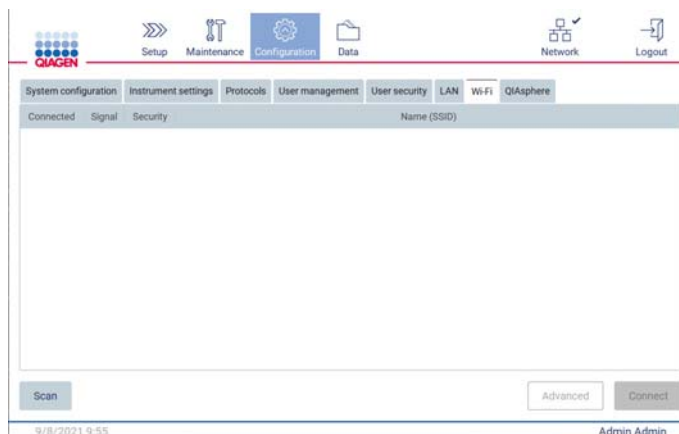


Picture 4.2-1 Touchscreen USB Ports

2. On the touchscreen, press the **Configuration** icon to enter the Configuration environment.



3. If the instrument is connected via the network cable at the back of the instrument, press the LAN tab header to switch to configuration of the Local Area Network (LAN).
 - 3.1. Set the LAN information according to the EZ2 site.
4. If the instrument is connected to the network using Wi-Fi, press the Wi-Fi tab header to switch to configuration of the Wi-Fi Network.
 - 4.1. Press the **Scan** button to scan for available Wi-Fi networks.
 - 4.2. From the list of available WiFi networks, select the one to connect the EZ2 to and press **Connect**.



Picture 4.2-2 Wi-Fi Configuration Screen

4.3. Enter the network password or other settings.

4.2.3 Camera Exposure Calibration

The camera exposure calibration must be performed on the EZ2 Connect Fx and EZ2 Connect MDx systems before the instrument can be used. If the instrument is relocated or moved, the camera exposure calibration must be performed again. This calibration is available to the clients and described in the User Manual. This is not applicable for the EZ2 Connect LS, as there is no camera on the EZ2 Connect LS.

1. To start the exposure calibration of the camera, press the **Maintenance** environment icon, then select the **Camera LED** tab.



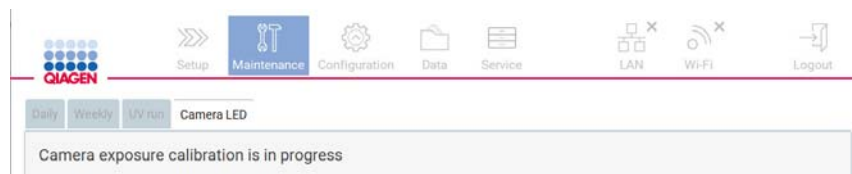
Picture 4.2-3 Camera exposure calibration

2. Follow the instructions to load the appropriate items into the EZ2 instrument.
 - Insert empty left and right tip racks into the appropriate location in the instrument.
3. Press the **Start** button after the instrument is correctly loaded and the hood closed.



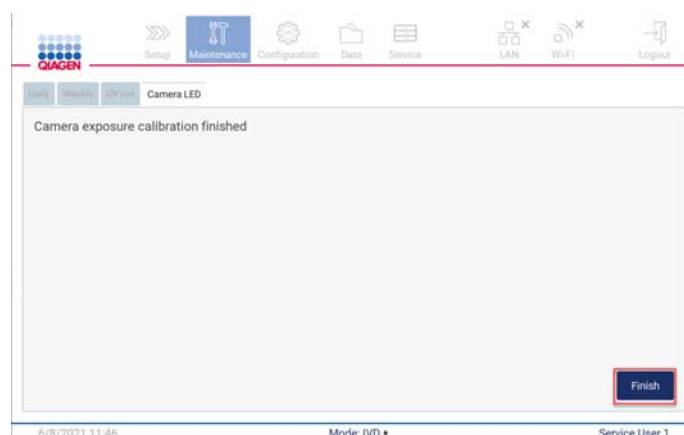
Picture 4.2-4 Press the Start button

4. During the calibration routine, the following will be displayed. Allow the calibration routine to complete.



Picture 4.2-5 Calibration in progress

5. Upon successful completion, the software will state the calibration is finished.
6. Once completed click the **Finish** button.

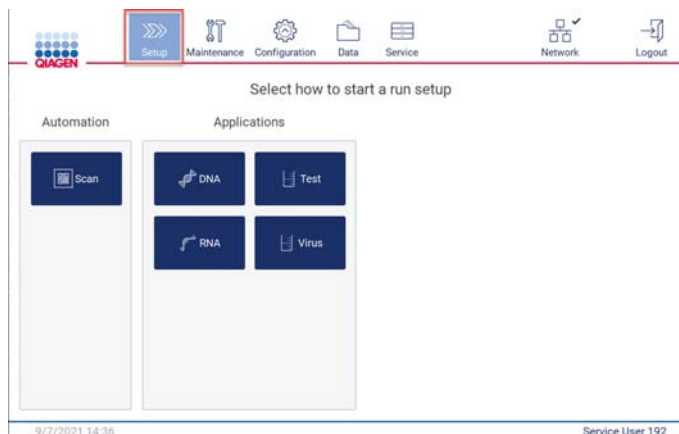


Picture 4.2-6 Camera exposure calibration finished

4.2.4 Workflow Installation Test

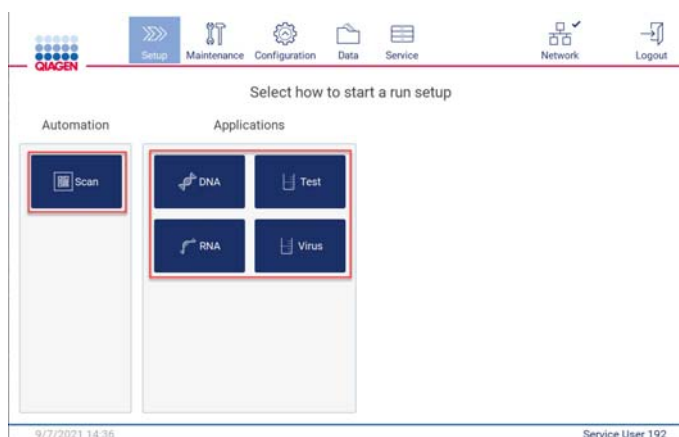
The workflow test incorporates a test of various components of the instrument, including the load check, the camera, and barcode reader.

1. Press the Setup environment icon.



Picture 4.2-7 Start a protocol

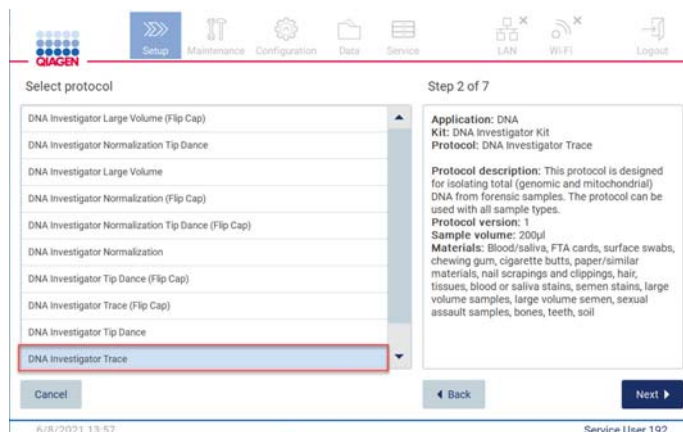
2. Press the **Scan** button under Automation to use the handheld barcode reader to scan an available Q-card with the kit to start a customer protocol.
 2.1. If no handheld barcode reader is available, press the appropriate Application button and select the appropriate kit.



Picture 4.2-8 Start a protocol for Fx and MDx instruments

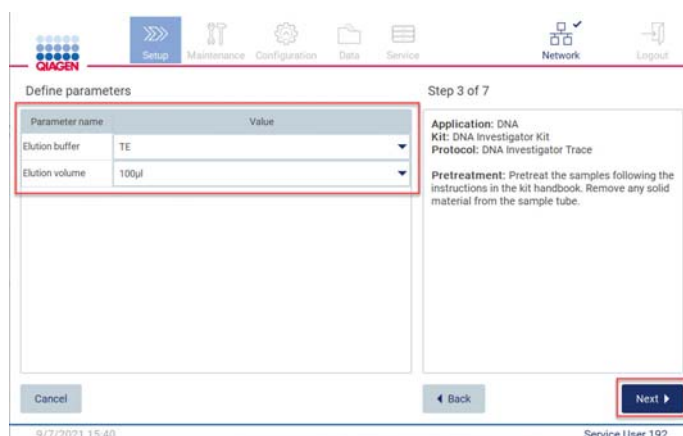
NOTE: 	The input tube and rack are specified in the Select protocol screen. Check that the Flip Cap or Standard rack, and thus tubes, are selected correctly.	
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3. Select a protocol from the available protocols for the kit. An example is shown below of the Select protocol screen. After protocol selection, click on **Next**.



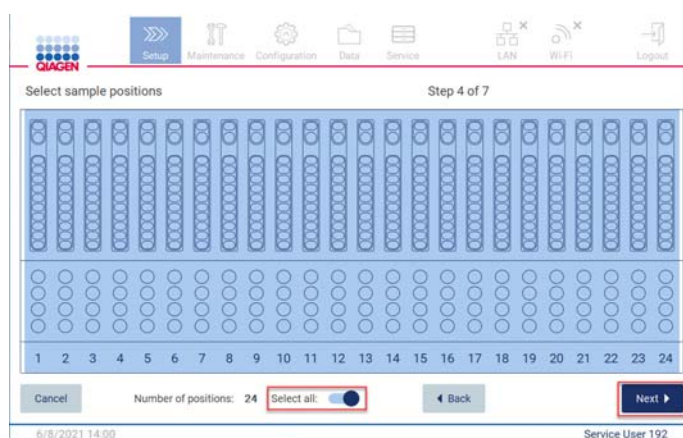
Picture 4.2-9 Select protocol

4. On the Define parameters screen, use the drop-down arrow to select the appropriate elution buffer and appropriate elution volume.
5. Press **Next** to continue.



Picture 4.2-10 Define parameters screen

6. Select all twenty-four positions using the Select all toggle.
 - Positions that will be used are indicated in blue.
7. Press **Next** when the appropriate positions are highlighted blue.
 - At least one position has to be selected for the Next button to be available.



Picture 4.2-11 Select positions that will be used

8. Automatically generate sample IDs by pressing the **Generate missing sample IDs** button.

Picture 4.2-12 Generate missing sample IDs button

9. Press **Next** after entering sample information.

Picture 4.2-13 Enter sample IDs Screen

10. For the EZ2 Connect Fx and EZ2 Connect MDx, check that the load check detects empty positions with the following steps.

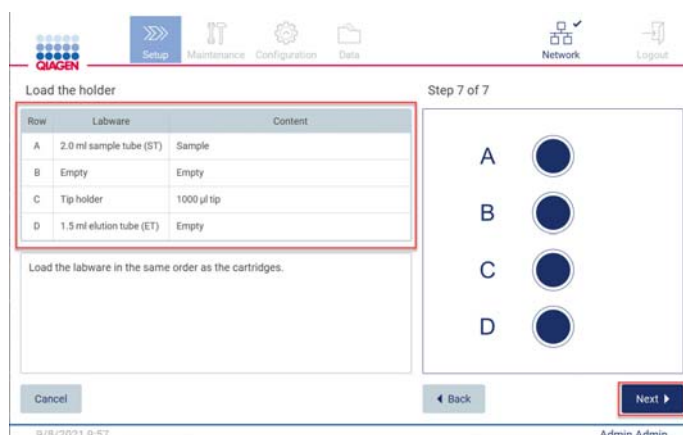
10.1. Load the empty reagent cartridge racks into the instrument.

10.2. Press Next once the empty reagent cartridge racks are loaded.

Picture 4.2-14 Load cartridge rack screen

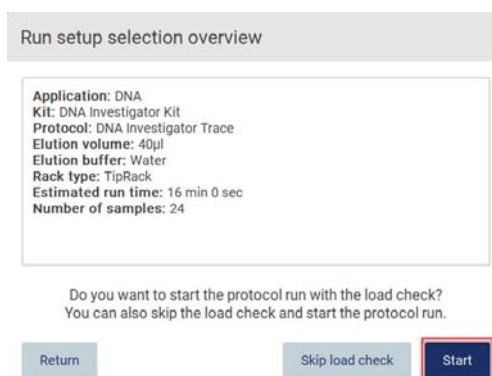
10.3. Load the empty tip racks into the instrument.

10.4. Press **Next** when the rack is loaded.

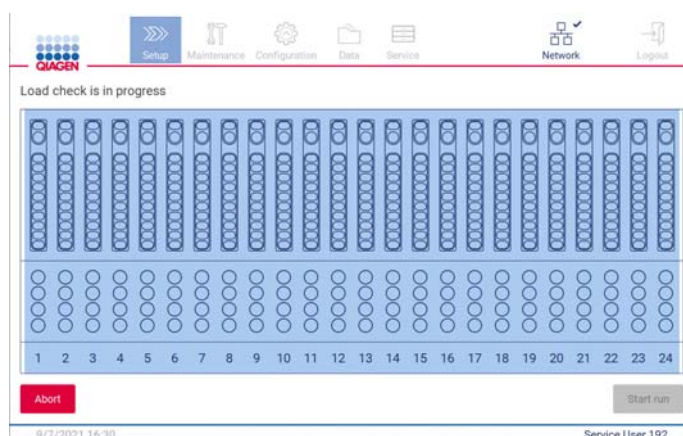


Picture 4.2-15 Load the holder screen

10.5. Press **Start** to begin the load check.



Picture 4.2-16 Confirm protocol and loading



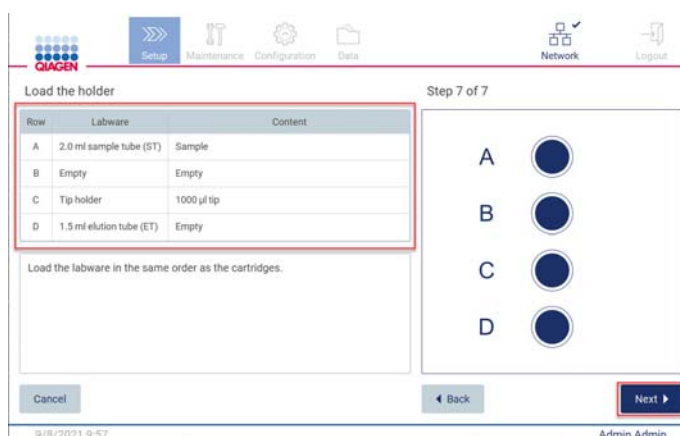
Picture 4.2-17 Load check in progress

10.6. The load check must fail due to the empty racks. The failed positions will be marked in red. Ensure all positions on the reagent rack and the tip rack that should not be empty are marked in red.



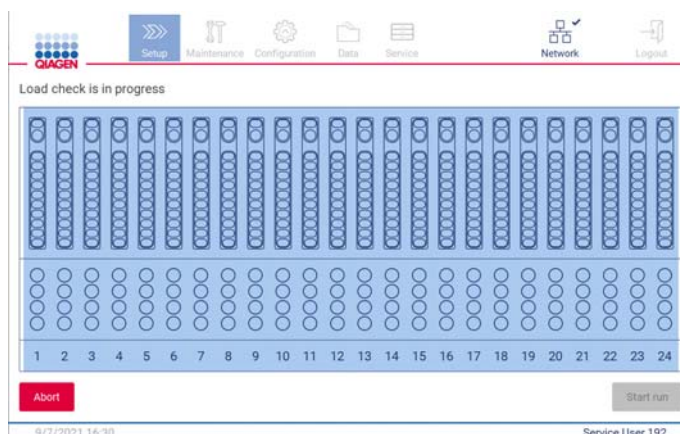
Picture 4.2-18 Load failed

- 10.7. After confirming the failed load check, press the **Back** button to setup the workdeck correctly.
11. Invert the reagent cartridges according to the handbook to resuspend the beads.
 12. Load twenty-four reagent cartridges in the reagent racks.
 13. If necessary for the protocol, load the tube in the heater block position as described in the GUI.
 14. Load the prepared reagent cartridge racks into the instrument.
 15. Load the tip racks with the appropriate consumables as shown on the Load consumables window.



Picture 4.2-19 Load the tip racks correctly

16. Load the prepared tip racks into the instrument.
17. Press **Start** to perform the load check again.



Picture 4.2-20 Load check in progress screen

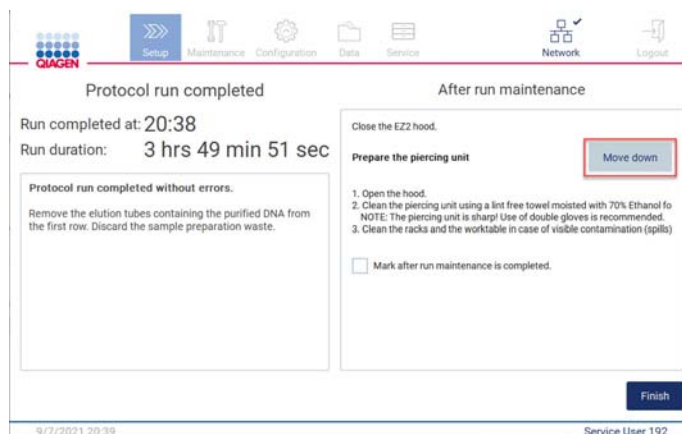
18. Confirm the load check finishes successfully when the instrument is correctly loaded.
19. Observe the protocol while running that:
 - 19.1. Aspiration of liquid is consistent in all pipette tips.
 - 19.2. Beads are aspirated and transferred.
 - 19.3. No drips occurred.
 - 19.4. Samples are processed correctly.
20. After completion of the run, check the sample volume remaining in the tube is consistent across all tubes and no beads or material entered the filter.
21. The test is successful if:
 - 21.1. When empty racks are loaded, the load check fails for the reagent racks, sample tubes, tips and elution tubes positions that should not be empty.
 - 21.2. Load check is successfully completed with the correctly loaded consumables.
 - 21.3. Tip pick up is smooth.
 - 21.4. The run completes successfully and without error.
 - 21.5. The LED strip is blinking.

4.2.5 After Run Maintenance

After each run is completed on the EZ2, the piercing unit must be cleaned.

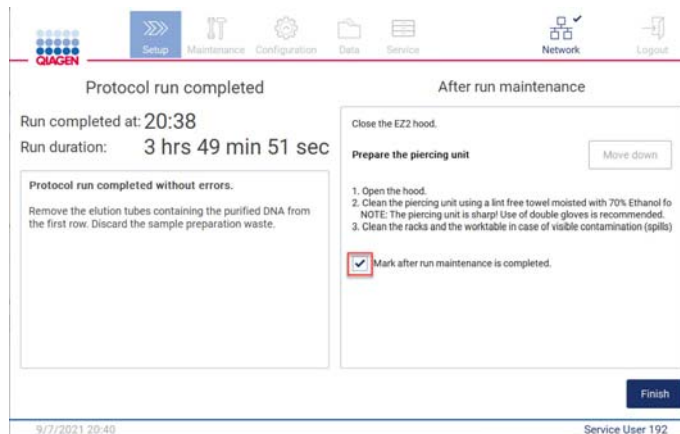
WARNING 	The piercing pins are sharp. Wear at least one pair of gloves and use caution.	
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1. After the run completes, press the **Move down** button under After run maintenance.
2. Once the piercing unit has moved into position, open the hood.
3. Moisten a lint-free cloth with 70% Ethanol to clean each of the piercing pins.



Picture 4.2-21 Perform after run maintenance

4. Check the **Mark after run maintenance is completed** box once the piercing unit is completely clean.

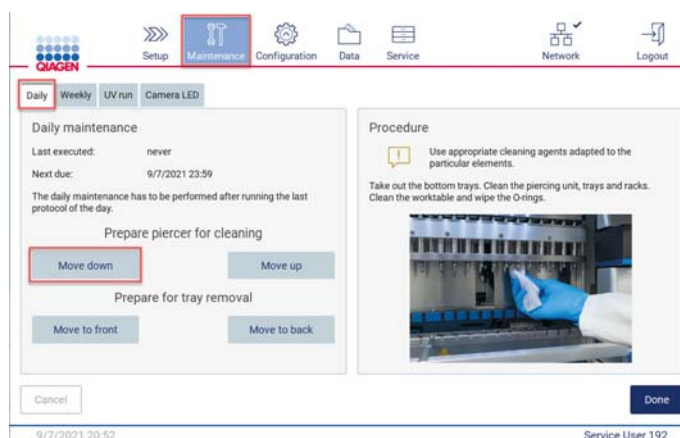


Picture 4.2-22 Mark after run maintenance as complete

5. With the mark after run maintenance box checked, press **Finish** to finalize the run.

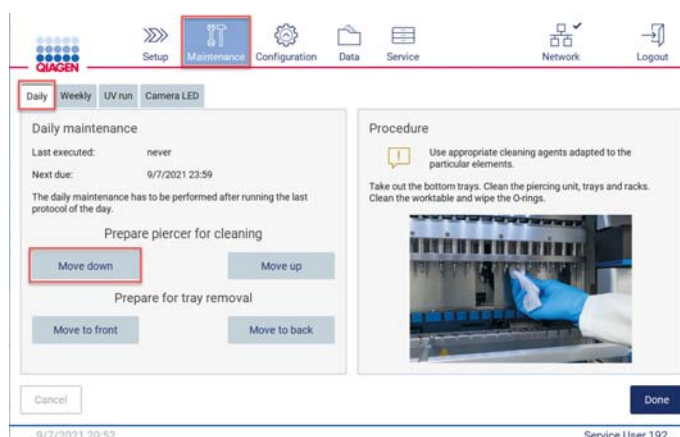
4.2.6 Daily Maintenance

1. To perform the Daily Maintenance, press the **Maintenance** environment icon.
2. If necessary, press the **Daily** tab.



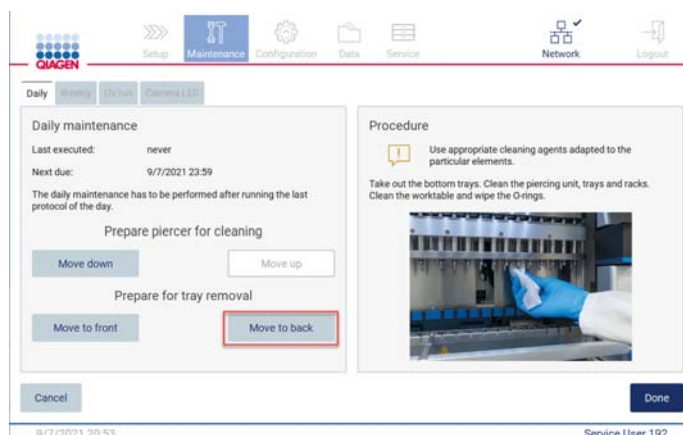
Picture 4.2-23 Start the daily maintenance

3. If the piercing unit was not already cleaned, press **Move down** under Prepare piercer for cleaning and clean the piercing pins with 70% Ethanol.
4. After the piercing unit is cleaned, press **Move up**.



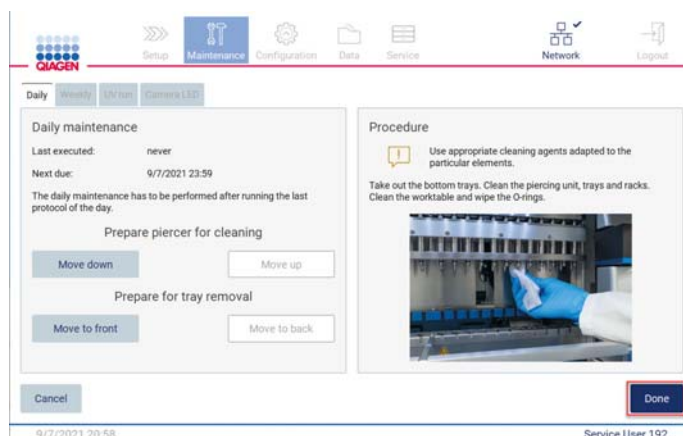
Picture 4.2-24 Move piercing unit up

- Remove the trays from the inside bottom of the instrument, by first moving the worktable by pressing **Move to back**.



Picture 4.2-25 Move worktable to the back of the instrument

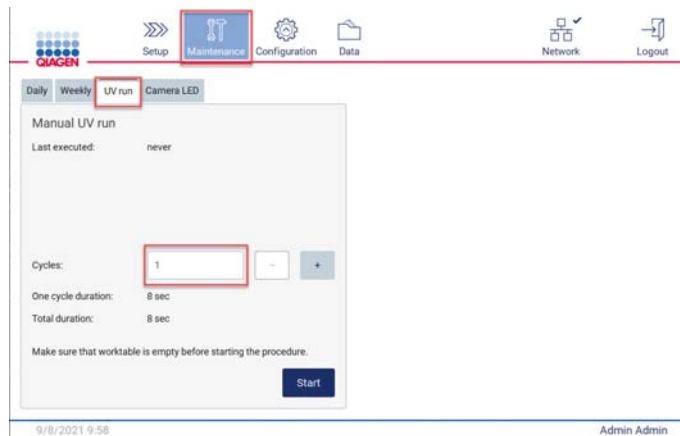
- Using the handle, lift the right tray gently out of the bottom of the instrument.
- After the right tray has been removed, lift the left tray gently out of the bottom of the instrument.
- Clean the trays.
- Re-install the left tray and then the right tray into the bottom of the instrument.
- After both trays are reinstalled, close the hood and press **Done**.



Picture 4.2-26 Finish daily maintenance

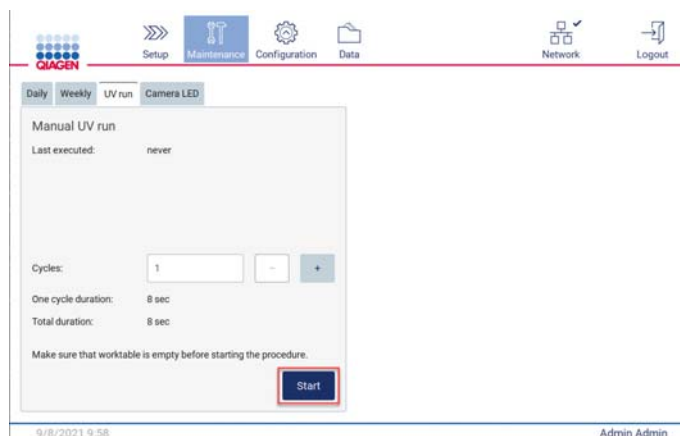
4.2.7 UV Test

- Press the **Maintenance** environment icon then the **UV Run** tab header.



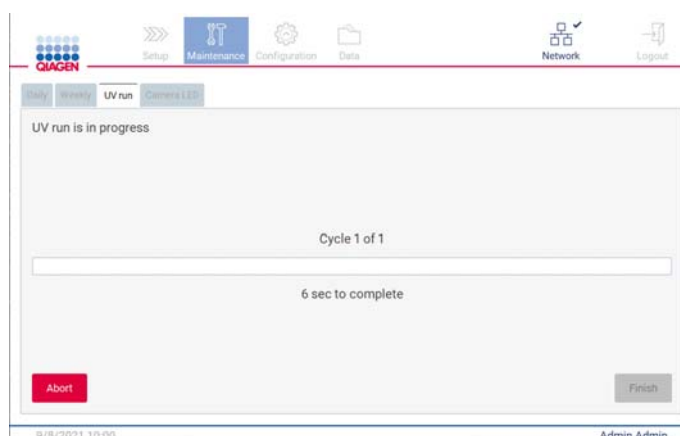
Picture 4.2-27 UV Run Screen

2. Confirm the cycles number is set to 1 cycle.
3. Confirm the worktable is empty; remove any plasticware as needed.
4. Press the **Start** button to start the UV cycle.



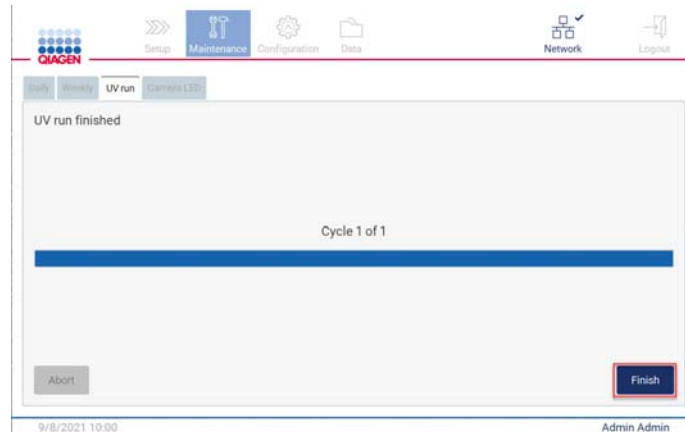
Picture 4.2-28 Start the UV Run

5. Allow the UV run to continue for 10 minutes. After at least 10 minutes, press the **Abort** button to stop the UV run.



Picture 4.2-29 UV Run progress

6. Confirm that the UV cycle should be aborted and wait for the instrument to home all modules.
7. Once the instrument has finish initializing, press **Finish** to complete the run.

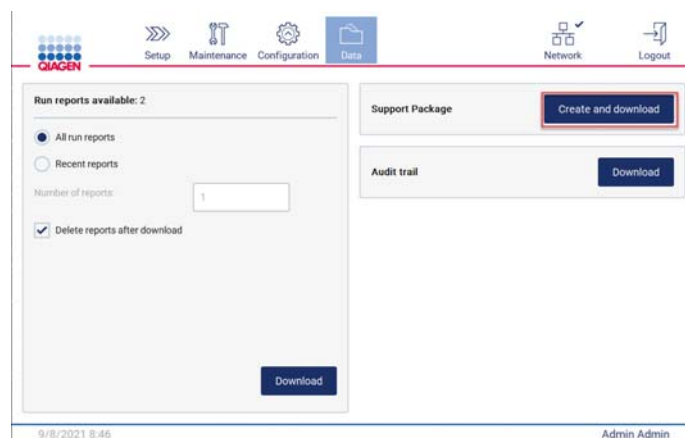


Picture 4.2-30 UV run finished

8. The UV run should not give any errors and movement should be smooth and consistent.

4.2.8 Installation Completion

1. Complete the Installation Checksheet (ICS-5000-0025), ensuring that all tasks have successfully been completed.
2. Download the Support Package from the instrument.
 - 2.1. Press **Data** to enter the Data environment.
 - 2.2. Press **Create and Download** next to Support Package to save the support package to the USB drive.



Picture 4.2-31 Create and download support package as Admin

3. Upload the completed Installation Checksheet (ICS-5000-0025), as well as the support package to Salesforce (SFDC) or CRM.
4. Mark any ISI notes that were implemented during the installation as complete in SFDC or CRM.
5. Close the work order or call according to the appropriate procedure.